

**Vidya Vikas Institute of Engineering and Technology,
Mysore
Library and Information Centre**



VTU Question Papers

MASTER OF COMPUTER APPLICATION

Odd Semester

2022-23

LIBRARY AND INFORMATION CENTRE

**Vidya Vikas Institute of Engineering and Technology
Bannur road, Alanahalli, Mysore-570028**

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CBCS SCHEME

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20MCA31

Third Semester MCA Degree Examination, Jan./Feb. 2023 Data Analytics using Python

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define keywords, statements, expressions, variables, precedence and associativity with examples and syntax. (10 Marks)
- b. Explain with syntax and example different types of Python data types and type() function. (10 Marks)

OR

- 2 a. Discuss different forms of if control statements with necessary examples. (10 Marks)
- b. What is a function? Mention its types. Write a python program to add two numbers using function, read input from the user. (10 Marks)

Module-2

- 3 a. Define string. Explain with necessary coding five basic string operations. Explain string slicing and joining. (10 Marks)
- b. Explain List creation, indexing and built in functions used on lists with syntax and examples. (10 Marks)

OR

- 4 a. Differentiate between sets, tuples and dictionaries. Write a python program to demonstrate encapsulation and overloading. (10 Marks)
- b. What is inheritance? Explain different types of inheritance with necessary example. (10 Marks)

Module-3

- 5 a. Define creating an array from Python lists. Explain numpy array attributes. (06 Marks)
- b. Discuss with example numpy array concatenation and splitting. (08 Marks)
- c. Explain specialized universal functions:
(i) Trigonometric (ii) Exponents and logarithms with necessary coding. (06 Marks)

OR

- 6 a. Mention Pandas data structures. Create a dataframe with three dimensional list state, year, POP (dictionary). Write necessary coding for retrieving row values and modifying column values. (06 Marks)
- b. Explain with example the concept of reindexing and ffill method. (06 Marks)
- c. How do we handle missing data in Python using Pandas? Explain with coding. (08 Marks)

Module-4

- 7 a. Explain reading and writing data in text format in Python with examples. (10 Marks)
- b. Explain the following methods with respect to database interaction:
i) Create ii) insert iii) connect iv) execute v) fetch all. (10 Marks)

OR

- 8 a. Explain with example the following merge methods:

i) inner ii) left iii) right

Create two dataframes with the following :

df1:

data1	key
0	b
1	b
2	a
3	c
4	a
5	b

df2:

data2	key
0	a
1	b
2	a
3	b
4	d

(10 Marks)

- b. Explain Data transforming using a function or mapping. Create a dataframe with the following columns:

Food	Ounce
Bacon	4.0
Pulled pork	3.0
Bacon	12.0
Honeyham	5.0

Add a column indicating the type of animal that each food come from.

(10 Marks)

Module-5

- 9 a. Write a Python program to plot sinusoid and cosine waves using Matplotlib and label them with necessary title and labels. (10 Marks)
- b. Explain with necessary coding creating a basic error bars and continuous errors. (10 Marks)

OR

- 10 a. Write a Python program to plot the histogram as follows (customized histogram). [Refer Fig.Q10(a)].

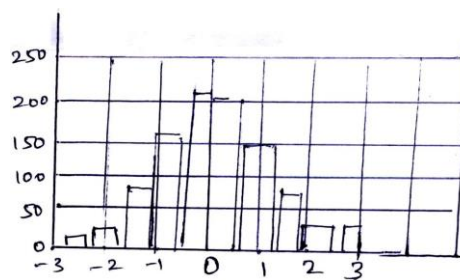


Fig.Q10(a)

(10 Marks)

- b. What is Seaborn plot? Explain pair plots for 'iris' dataset and kernel density estimation using kdeplot and displot. (10 Marks)

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20MCA32

Third Semester MCA Degree Examination, Jan./Feb. 2023

Internet of Things

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1
 - a. Define IOT and discuss the genesis of IOT with evolutionary phases of internet. (04 Marks)
 - b. Describe IOT world Forum standardized architecture. (06 Marks)
 - c. Explain the IOT impacts in real world. (04 Marks)
 - d. Explain with neat diagram, the one M2M IOT standardized architecture. (06 Marks)

OR

- 2
 - a. Explain IOT data management and compute stack. (08 Marks)
 - b. Explain core IOT functional stack. (08 Marks)
 - c. Write the different challenges of IOT. (04 Marks)

Module-2

- 3
 - a. Explain briefly the Wireless Sensor Networks (WSN). (06 Marks)
 - b. Explain different communication criteria of IOT. (06 Marks)
 - c. What is Zigber? Explain 802.15.4 physical layer, MAC layer, and security. (08 Marks)

OR

- 4
 - a. Define sensors and actuators. Explain how they interact with the physical world. (06 Marks)
 - b. What is SANET? Explain some advantages and disadvantages that a wireless based solution offers. (06 Marks)
 - c. List and explain different types of sensors. (08 Marks)

Module-3

- 5
 - a. What are the key advantages of Internet Protocol? Explain each of them. (05 Marks)
 - b. Discuss the need for optimization. (05 Marks)
 - c. Describe application protocol for IOT. (10 Marks)

OR

- 6
 - a. Explain 6TiSCH in detail. (10 Marks)
 - b. Discuss various IOT application transport methods. (10 Marks)

Module-4

- 7
 - a. List the common challenges in OT security. Explain any 4 of them. (10 Marks)
 - b. Explain any 2 Bigdata analytics tools and technology. (10 Marks)

OR

- 8 a. Discuss the following:
i) Supervised learning
ii) Unsupervised learning
iii) Neural network. (06 Marks)
b. Explain OCTAVE and FAIR formal risk analysis. (06 Marks)
c. Explain edge streaming analytics in detail. (08 Marks)

Module-5

- 9 a. What is Arduino? What are the advantages of Arduino? (05 Marks)
b. Explain the following with respect to the fundamentals of arduino programming:
i) Structure
ii) Function
iii) Variables and data types
iv) Flow control statements. (10 Marks)
c. Explain IOT strategy for smart cities. (05 Marks)

OR

- 10 a. With a neat diagram, explain Raspberry Pi board and its connections. (10 Marks)
b. Explain smart city IOT architecture. (10 Marks)

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Third Semester MCA Degree Examination, Jan./Feb. 2023

Advances in Java

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Servlet. Explain life cycle of a Servlet with a neat diagram. (10 Marks)
- b. Define Cookies. Write a JAVA Servlet program using Cookies to remember user preferences. (10 Marks)

OR

- 2 a. Explain 4 different ways in which a session can be tracked with suitable examples. (10 Marks)
- b. Briefly explain any 5 HTTP status codes during server responses. (05 Marks)
- c. Explain HTTP Request headers in brief. (05 Marks)

Module-2

- 3 a. Explain different kinds of main tags used in JSP with example for each. (10 Marks)
- b. Explain different ways in which you can invoke JAVA code from JSP. (10 Marks)

OR

- 4 a. Explain the need, benefits and advantages of JSP over competing technologies. (10 Marks)
- b. Write a JSP program that adds two numbers entered through HTML form and display the result (10 Marks)

Module-3

- 5 a. List all the attributes of Page directives tags in JSP and explain any five with an example for each. (10 Marks)
- b. Write a JSP program to get student information through HTML form. Create a JAVA Bean class, populate Bean and display the same information through another JSP. (10 Marks)

OR

- 6 a. Define JAVA Bean and state its advantages. Explain the features of JAVA Bean. (08 Marks)
- b. Write a note on JAR files. (06 Marks)
- c. Write a JSP program to include an applet along with necessary applet code. (06 Marks)

Module-4

- 7 a. With an example, explain the essential steps of JDBC program. (10 Marks)
- b. Explain in brief basic and advanced JDBC data types. (10 Marks)

OR

- 8 a. What are annotations? Discuss built-in annotations with an example. (10 Marks)
b. Explain the types of statements objects in JDBC with an example. (10 Marks)

Module-5

- 9 a. Explain the following terms of container services in EJB:
(i) Transactions
(ii) Security.
(iii) Naming and Object stores (10 Marks)
b. What is stateless session bean? Explain the life cycle of stateless session bean with a neat diagram. (10 Marks)

OR

- 10 a. What is message driven bean? Explain the life cycle of message driven bean. (10 Marks)
b. Write short notes on :
(i) Stateful session bean.
(ii) Singleton session bean. (10 Marks)

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20MCA344

Third Semester MCA Degree Examination, Jan./Feb. 2023 Software Testing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain error, faults and failures in the process of programming and testing with a diagram. (10 Marks)
 b. Describe the quality attributes of testing. (06 Marks)
 c. Differentiate between testing and debugging. (04 Marks)

OR

- 2 a. Discuss the different types of test metrics. (10 Marks)
 b. What are the principles for analysis and testing? Explain in detail. (10 Marks)

Module-2

- 3 a. With the help of diagram, explain testing lifecycle. (08 Marks)
 b. Discuss specified, implemented and tested behaviour of software with the help of Venn diagram. (08 Marks)
 c. Discuss fault taxonomy and give two examples of each type. (04 Marks)

OR

- 4 a. Write a pseudo – code for triangle problem with data flow diagrams. (10 Marks)
 b. Write the pseudo – code for the implementation of the NextDate function. (10 Marks)

Module-3

- 5 a. Explain the boundary value analysis and generalizing boundary value analysis. (08 Marks)
 b. What is boundary value analysis? Discuss the following testing with examples.
 i) Robustness testing
 ii) Worst case testing. (08 Marks)
 c. Write the testcases for NEXTDate function using boundary value analysis. (04 Marks)

OR

- 6 a. Write the testcases for the commission problem using equivalence class testing. (10 Marks)
 b. Represent the triangle problem in a decision table format. (06 Marks)
 c. Define the following :
 i) Boundary value test
 ii) Equivalence class test
 iii) Random test
 iv) Decision table test. (04 Marks)

(04 Marks)

Module-4

- 7 a. Draw program graph and DD path graph for a triangle problem. (10 Marks)
b. Explain McCabe's basis path method with an example. (10 Marks)

OR

- 8 a. Discuss waterfall spin-offs model and system testing briefly. (06 Marks)
b. Explain traditional view of testing levels and alternative lifecycle models. (08 Marks)
c. Draw the context diagram of SATM and discuss the type of testing for which it can be used. (06 Marks)

Module-5

- 9 a. Explain mutation analysis with its variants. (10 Marks)
b. What is scaffolding and what is the purpose of it? Differentiate generic and specific scaffold (10 Marks)

OR

- 10 a. Brief about test oracle and self check as oracles. (04 Marks)
b. Explain any five major activities in planning and monitoring the process. (10 Marks)
c. Discuss the different types of risks involved in process management and quality management. (06 Marks)

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20MCA354

Third Semester MCA Degree Examination, Jan./Feb. 2023

Software Project Management

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is Project? Explain the characteristics of the project and how software projects are different from other types of projects. (10 Marks)
- b. What are the activities covered by Software Project Management? Explain. (10 Marks)

OR

- 2 a. With neat diagram, explain Project Control Cycle. (10 Marks)
- b. Explain some ways of categorizing Software projects. (10 Marks)

Module-2

- 3 a. Explain different methods of evaluating Individual projects. (10 Marks)
- b. Discuss the different cost – benefit of Evaluation techniques. (10 Marks)

OR

- 4 a. What is Financial Accounting? Explain the principles and standard. (10 Marks)
- b. Explain the various Project Risk Evaluation with example. (10 Marks)

Module-3

- 5 a. Explain the objectives of Activity Planning. (10 Marks)
- b. List and explain the three approaches used for identifying activities or tasks that make up a project. (10 Marks)

OR

- 6 a. List out the Activity on Arrow rules and Conventions with example. (10 Marks)
- b. Explain a framework for dealing with a Risk. (10 Marks)

Module-4

- 7 a. Explain the different Visualizing methods for a Project Progress. (10 Marks)
- b. Write and explain flow chart of Project Control Cycle. (10 Marks)

OR

- 8 a. Explain in detail the Review process model with neat diagram. (10 Marks)
- b. Explain how the project can take back to target. (10 Marks)

Module-5

- 9 a. Explain Oldman Hackman Job characteristics model. (08 Marks)
- b. Write a note on :
 - i) Team Formation model.
 - ii) Individual characteristics to need of balanced team. (12 Marks)

OR

- 10 a. Explain the Maslow's hierarchy of needs. (05 Marks)
- b. Write a note on Leadership. (05 Marks)
- c. Explain in detail decision making in Software Project Management. (10 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg, 42+8= 50, will be treated as malpractice.

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First Semester MCA Degree Examination, Jan./Feb. 2023
Mathematical Foundation for Computer Applications

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Define, cardinality of a set, singleton set and universal set with example.	6	L2	CO1
	b.	Define union and intersection of two sets with example.	4	L2	CO1
	c.	Find the Eigen values and Eigen vectors of the matrix $A = \begin{bmatrix} 7 & 3 \\ 3 & -1 \end{bmatrix}$.	10	L2	CO1
OR					
Q.2	a.	Define matrix. Explain different types of matrices with example.	8	L2	CO1
	b.	Let $A = \{1, 2, 3, 4, 5, 6\}$, $B = \{6, 7, 8, 9, 10\}$ and $f : A \rightarrow B$ be a function defined by $f = \{(1, 7)(2, 7)(3, 8)(4, 6)(5, 9)(6, 9)\}$. Determine $f^{-1}(6)$ and $f^{-1}(9)$. Also if $B_1 = \{7, 8\}$, $B_2 = \{8, 9, 10\}$ then find $f^{-1}(B_1)$ and $f^{-1}(B_2)$.	4	L1	CO1
	c.	In a class of 52 students, 30 are studying C++, 28 are studying pascal and 13 are studying both languages. How many in this class are studying at least one of these languages? How many are studying neither of these languages?	6	L2	CO1
	d.	State and explain Pigeon hole principle.	2	L1	CO1
Module – 2					
Q.3	a.	State the laws of logic.	8	L2	CO2
	b.	Write the contra positive, converse and the inverse of the conditional statement. “If oxygen is a gas then Gold is compound”.	6	L1	CO2
	c.	Define Tautology. Show that the compound proposition, $[P \rightarrow (q \rightarrow r)] \rightarrow [(p \rightarrow q) \rightarrow (p \rightarrow r)]$ is a Tautology.	6	L3	CO2
OR					
Q.4	a.	Prove the following is valid argument : $\begin{array}{l} p \rightarrow r \\ \neg p \rightarrow q \\ q \rightarrow s \\ \hline \therefore \neg r \rightarrow s \end{array}$	8	L2	CO2

	b.	Give the direct proof of the following statement “If n is an odd integer, then n^2 is odd”.	6	L2	CO2																
	c.	What is a proposition? Let p and q be the propositions “Swimming in the New Jersey sea shore is allowed and sharks have been near the sea shore.” Express each of the following compound propositions as an English sentence. (i) $p \rightarrow \sim q$ (ii) $\sim p \rightarrow \sim q$ (iii) $p \leftrightarrow q$	6	L1	CO2																
Module – 3																					
Q.5	a.	Let $A = \{1, 2, 3, 4\}$, $R = \{(1, 3)(1, 1)(3, 1)(1, 2)(3, 3)(4, 4)\}$ be a relation on A . Determine whether R is reflexive, symmetric, asymmetric and write matrix representation.	6	L2	CO3																
	b.	If $A = \{1, 2, 3, 4\}$ and $R = \{(1, 2)(1, 3)(2, 4)(4, 4)\}$, $S = \{(1, 1)(1, 2)(1, 3)(1, 4)(2, 3)(2, 4)\}$ be relations on A then find RoS , SoR , R^2 and S^2 . Also write their matrices.	8	L2	CO3																
	c.	Discuss briefly on partitions and equivalence classes with example.	6	L2	CO3																
OR																					
Q.6	a.	Show that the set $A = \{1, 2, 3, 4, 6, 8, 12\}$ is a POSET with respect to the relation R defined as $\{(a, b) : a \text{ divides } b\}$ and draw its Hasse diagram.	8	L3	CO3																
	b.	Draw the directed graph of relation, $R = \{(1, 1)(1, 3)(2, 1)(2, 3)(2, 4)(3, 1)(3, 2)(4, 1)\}$ on the set $\{1, 2, 3, 4\}$. Also find in-degree and out-degree of each vertex.	6	L2	CO3																
	c.	Define lattices. Determine whether the POSET $(\{1, 2, 3, 4, 5\}, 1)$ is lattice or not.	6	L2	CO3																
Module – 4																					
Q.7	a.	A random variable X has the following probability distribution. <table border="1"><tr><td>X</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td></tr><tr><td>$P(X)$</td><td>K</td><td>$3K$</td><td>$5K$</td><td>$7K$</td><td>$9K$</td><td>$11K$</td><td>$13K$</td></tr></table> (i) Find K . (ii) Evaluate $P(X < 4)$, $P(X \geq 5)$, $P(3 < X \leq 6)$. (iii) Find the minimum value of K so that $P(X \leq 2) > 0.3$	X	0	1	2	3	4	5	6	$P(X)$	K	$3K$	$5K$	$7K$	$9K$	$11K$	$13K$	10	L2	CO4
X	0	1	2	3	4	5	6														
$P(X)$	K	$3K$	$5K$	$7K$	$9K$	$11K$	$13K$														
	b.	The probability that a pen manufactured by a company will be defective is $\frac{1}{10}$. If 12 such pens are manufactured, find the probability that, (i) Exactly two will be defective (ii) at least two will be defective (iii) none will be defective.	10	L2	CO4																
OR																					
Q.8	a.	Given that 2% of the fuses manufactured by a firm are defective. Find by using Poisson distribution, the probability that a box containing 200 fuses has (i) no defective fuses (ii) 3 or more defective fuses (iii) at least one defective fuse.	8	L2	CO4																

	b. The length of a telephone conversation has an exponential distribution with a mean of 3 minutes. Find the probability that a call (i) Ends in less than 3 minutes. (ii) Takes between 3 and 5 minutes.	6	L2	CO4
	c. Find the constant C such that, $f(x) = \begin{cases} Cx^2, & 0 < x < 3 \\ 0, & \text{otherwise} \end{cases}$ is a probability density function. Also compute $P(1 < X < 2)$.	6	L2	CO4

Module – 5

Q.9	a. Define the following with suitable examples: (i) Simple graph (ii) Complete graph (iii) Bipartite graph	6	L2	CO5
	b. Verify the following graphs are isomorphic or not. Fig.Q9 (b)	6	L2	CO5
	c. Explain the Konigsberg bridge problem.	8	L1	CO5

OR

Q.10	a. Determine $ V $ for the graph $G = (V, E)$ if G has 10 edges with two vertices of degree 4 and others of degree 3.	4	L2	CO6
	b. Define the following with suitable examples : (i) Euler's graph (ii) Hamilton graph	6	L2	CO6
	c. Use Dijkstra's algorithm to find the length of a shortest path between the vertices a and z in the graph given below. 	10	L2	CO6

Fig. Q10 (c)

First Semester MCA Degree Examination, Jan./Feb. 2023

Operating System Concepts

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C															
Q.1	a.	What is an operating system? Explain the various services of the operating system.	10	L1	CO1															
	b.	What is a system call? Explain the different types of system calls.	10	L1	CO1															
OR																				
Q.2	a.	Explain simple, layered and microkernel structures of the operating system.	10	L1	CO1															
	b.	What are Virtual Machines? Explain the implementation of virtual machines.	10	L1	CO1															
Module – 2																				
Q.3	a.	What is a process? Explain the five state process model with a neat diagram.	10	L1	CO1															
	b.	Consider the following processes, which have arrived at the ready queue with the burst and the arrival time given in milliseconds as shown below: <table border="1"><thead><tr><th>Process</th><th>Burst Time</th><th>Arrival Time</th></tr></thead><tbody><tr><td>P1</td><td>8</td><td>0</td></tr><tr><td>P2</td><td>4</td><td>1</td></tr><tr><td>P3</td><td>9</td><td>2</td></tr><tr><td>P4</td><td>5</td><td>3</td></tr></tbody></table> Draw the Gantt chart and calculate the average waiting time using the following scheduling algorithms: (i) FCFS (ii) SJF (Preemptive) (iii) RR (Q = 4)	Process	Burst Time	Arrival Time	P1	8	0	P2	4	1	P3	9	2	P4	5	3	10	L3	CO5
Process	Burst Time	Arrival Time																		
P1	8	0																		
P2	4	1																		
P3	9	2																		
P4	5	3																		
OR																				
Q.4	a.	What is a process control block? Explain the use of PCB in context switching.	10	L2	CO1															
	b.	What are user threads and kernel threads? Explain the various multithreading models.	10	L2	CO1															
Module – 3																				
Q.5	a.	What is a critical section problem? Illustrate Peterson’s two process solution for a critical section problem.	10	L2	CO2															
	b.	What are Semaphores? Explain the producer-consumer problem and give a solution using semaphores.	10	L3	CO4															
OR																				

Q.6	a.	What is a deadlock? What are the necessary conditions for a deadlock to occur?	10	L2	CO2																																																																						
	b.	Consider the following snapshot of a system: <table><tr><td></td><td colspan="3">Allocation</td><td colspan="3">Max</td><td colspan="3">Available</td></tr><tr><td></td><td>A</td><td>B</td><td>C</td><td>A</td><td>B</td><td>C</td><td>A</td><td>B</td><td>C</td></tr><tr><td>P₀</td><td>0</td><td>1</td><td>0</td><td>7</td><td>5</td><td>3</td><td>3</td><td>3</td><td>2</td></tr><tr><td>P₁</td><td>2</td><td>0</td><td>0</td><td>3</td><td>2</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P₂</td><td>3</td><td>0</td><td>2</td><td>9</td><td>0</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P₃</td><td>2</td><td>1</td><td>1</td><td>2</td><td>2</td><td>2</td><td></td><td></td><td></td></tr><tr><td>P₄</td><td>0</td><td>0</td><td>2</td><td>4</td><td>3</td><td>3</td><td></td><td></td><td></td></tr></table> Answer the following questions using Bankers Algorithm: (i) Construct a need matrix. (ii) Is the system in a safe state? If yes, what is the safe sequence? (iii) If process P₁ makes a request (1, 0, 2), can the request be granted?		Allocation			Max			Available				A	B	C	A	B	C	A	B	C	P ₀	0	1	0	7	5	3	3	3	2	P ₁	2	0	0	3	2	2				P ₂	3	0	2	9	0	2				P ₃	2	1	1	2	2	2				P ₄	0	0	2	4	3	3				10	L3	CO5
	Allocation			Max			Available																																																																				
	A	B	C	A	B	C	A	B	C																																																																		
P ₀	0	1	0	7	5	3	3	3	2																																																																		
P ₁	2	0	0	3	2	2																																																																					
P ₂	3	0	2	9	0	2																																																																					
P ₃	2	1	1	2	2	2																																																																					
P ₄	0	0	2	4	3	3																																																																					
Module – 4																																																																											
Q.7	a.	Write a C program to simulate the multiprogramming with variable member of tasks (MVT) memory management technique. Given the size of the memory, size of OS, number of processes and size of each process, calculate the external fragmentation.	10	L3	CO5																																																																						
	b.	What is Paging? Explain the paging hardware with a neat diagram.	10	L2	CO5																																																																						
OR																																																																											
Q.8	a.	What is demand paging? Explain how demand paging can be implemented.	10	L2	CO5																																																																						
	b.	Consider the following reference string:7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1 7 0 1 How many page faults would occur in case of the following page replacement algorithms: (i) Optimal (ii) LRV? Assuming 3 frames. Note : Initially all frames are empty.	10	L3	CO3																																																																						
Module – 5																																																																											
Q.9	a.	What are the various access methods used for accessing files?	10	L2	CO5																																																																						
	b.	Explain the various directory structures with neat diagrams.	10	L2	CO5																																																																						
OR																																																																											
Q.10	a.	Write a note on different file allocation methods.	10	L2	CO5																																																																						
	b.	Show how free space management is done using: (i) Bit vector (ii) Linked list (iii) Grouping (iv) Counting	10	L3	CO5																																																																						

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First Semester MCA Degree Examination, Jan./Feb. 2023

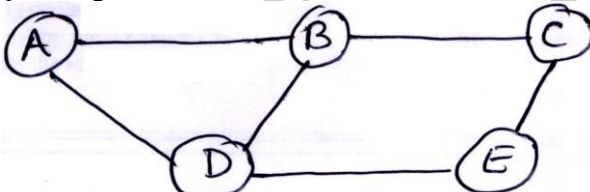
Data Structures

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.*

Module – 1				M	L	C
Q.1	a.	Define data structures. Explain classification of data structures with example.	10	L2	CO1	
	b.	Explain the two types of representation of a stack with neat diagram.	4	L2	CO1	
	c.	Write an algorithm to convert a valid infix expression into postfix expression.	6	L3	CO2	
OR						
Q.2	a.	What is stack? Explain the stack operations push and POP with C functions.	8	L3	CO1	
	b.	Convert the following expression from infix to postfix: i) $(A + B) * C - D \wedge E$ ii) $A + B / (D + E) * F \wedge G$	4	L3	CO2	
	c.	Write an algorithm to evaluate post fix expression using stack.	8	L3	CO2	
Module – 2						
Q.3	a.	What is recursion? Write C program to find GCD of 3 Nos using recursion.	8	L3	CO2	
	b.	What is Queue? Write C program to implement queue operations Enqueue, Dequeue and display using array as datastructure.	12	L3	CO3	
OR						
Q.4	a.	Explain queue variants with neat diagram.	4	L2	CO3	
	b.	Implement circular queue operations Insert, Remove and Display using C program.	10	L3	CO3	
	c.	Write a C program to find factorial of a number using recursion.	6	L3	CO3	
Module – 3						
Q.5	a.	What is DMA? Discuss different types of memory management functions in C.	10	L2	CO2	
	b.	Write functions in C to demonstrate the following operations on singly linked list: i) Insert at front ii) Remove node from front iii) Insert node at end.	10	L3	CO3	

OR					
Q.6	a.	Explain getNode() and freeNode() operation with C code snippet.	6	L2	CO3
	b.	Write a note on header nodes in linked list with neat diagram.	4	L2	CO3
	c.	Write C code snippet to demonstrate application of linked list as stack for implementing push and POP operations.	10	L3	CO3
Module – 4					
Q.7	a.	What is binary tree? Write note on threaded binary tree with neat diagram.	10	L2	CO3
	b.	Construct the binary search tree for the data 27, 5, 36, 47, 19, 250, 21, 44, 6. Perform preorder, Inorder and Postorder traversal for the constructed BST.	10	L3	CO4
OR					
Q.8	a.	Write a note on Array and linked representation of binary tree.	8	L3	CO3
	b.	Write recursive function for the following operations on binary search tree: i) Insert key in BST ii) Search key in BST.	12	L3	CO3
Module – 5					
Q.9	a.	Define DFS and BFS. Obtain the difference between both. Traverse the following graph using DFS and BFS. Assume the source node is B.	10	L3	CO4
		 Fig.Q.9(a)			
	b.	What is radix sort? Sort the following numbers using radix sort and show the table of various passes of radix sort. List : 326, 453, 608, 835, 751, 435, 704, 690.	10	L3	CO4
OR					
Q.10	a.	What is a graph? Explain array representation of graph with neat diagram.	4	L2	CO4
	b.	What is hash collision? Explain various methods for resolving hash collisions.	6	L2	CO4
	c.	Implement the hash function $h(k) = k \% 10$ on the numbers: 10, 34, 23, 56, 14, 89, 65, 94, 53, 21, 63, 33. Show the hash table. Resolve hash clashes using separate chaining method.	10	L3	CO4

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First Semester MCA Degree Examination, Jan./Feb. 2023

Computer Networks

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Briefly explain the five components of Data Communication System with an example each.	05	L2	CO1
	b.	Discuss the functionalities of Physical, Data link and Network layers of OSI reference model with suitable diagrams.	10	L2	CO1
	c.	Explain the different types of addresses used in TCP/IP.	05	L2	CO1
OR					
Q.2	a.	Discuss the importance of Transport, Session and Presentation layers in OSI model.	10	L2	CO1
	b.	Explain TCP/IP protocol suite with a neat labeled diagram. Describe the transport layer protocols such as TCP and UDP.	10	L2	CO1
Module – 2					
Q.3	a.	Explain the characteristics of Analog and Digital signals.	06	L2	CO2
	b.	Discuss the causes for transmission impairment in computer networks.	08	L2	CO2
	c.	Compare the differences between Nyquist and Shannon's Theorem. A Transmission channel with 1-MHz bandwidth and the SNR for this channel is 63. Find the appropriate bit rate and signal level.	06	L3	CO2
OR					
Q.4	a.	Discuss the different phases of PCM in detail with a suitable example.	10	L2	CO2
	b.	Sketch the digital signal line coding for the given bit stream 01010000111101011 for Biphase and Bipolar encoding schemes. Analyse the characteristics of each output signal to select the best one.	10	L3	CO2
Module – 3					
Q.5	a.	Explain the process of multiplexing and demultiplexing with a neat labeled diagram.	06	L2	CO3
	b.	Define Virtual Circuit Identifier. Perform request and acknowledgement setup in virtual circuits with a neat sketch.	06	L3	CO3
	c.	Explain Frequency Hopping Spread Spectrum. Illustrate with an example.	08	L3	CO3
OR					
Q.6	a.	Make out the differences between Synchronous TDM and Statistical TDM.	06	L4	CO2
	b.	List out the Data Rate Management Techniques used in multiplexing, explain briefly. State the reason for synchronization during multiplexing.	08	L2	CO3
	c.	Mention the advantages and disadvantages of Circuit Switched Networks over Datagram Networks.	06	L2	CO3
1 of 2					

Module – 4

Q.7	a.	Explain the procedure employed for error detection and error correction in Block coding.	06	L2	CO4
	b.	Calculate the CRC for the given bit stream 1101011011 with given generator polynomial $G(x) = x^4 + x + 1$. (i) Verify the message at receiver side if bit stream remains. (ii) Inspect the error, if the third bit from the left is being inverted during the transmission.	10	L4	CO4
	c.	Recall the different types of errors occurred during data transmission in networks.	04	L1	CO4

OR

Q.8	a.	Discuss the design of the encoder and decoder of Hamming code generation. Explain the two dimensional parity check value.	10	L5	CO4
	b.	(i) Write the steps involved in Internet Checksum algorithm on both sender and receiver side. (ii) Evaluate checksum value of 1 0 0 1 0 0 1 1 1 0 0 1 0 0 1 1 and 1 0 0 1 1 0 0 0 0 1 0 0 1 1 0 1 considering 16-bit data segment.	10	L5	CO4

Module – 5

Q.9	a.	Differentiate between fixed and variable size framing. Explain character stuffing and bit stuffing process. Apply bit stuffing for the following data bit sequence. 1 0 1 0 0 1 1 1 1 1 0 0 0 0 0 0 1 1 1 1 1 1 0 0	10	L3	CO4
	b.	Discuss the design of stop and wait protocol for noiseless channel. Write an algorithm for the same on the sender and receiver side.	10	L3	CO4

OR

Q.10	a.	Give the taxonomy of protocols of Flow and Error control. Briefly explain Go-Back-N ARQ protocol with an example.	10	L3	CO4
	b.	Explain the frame formats of HDLC and PPP protocols in detail.	10	L3	CO4

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Design and Analysis of Algorithms

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Define algorithm. Illustrate the characteristics of an algorithm.	06	L1	CO1
	b.	List and explain the different problem types of an algorithm.	04	L1	CO1
	c.	Explain asymptotic notations, with a diagram and explain with an example.	10	L2	CO1
OR					
Q.2	a.	Design a general plan for analyzing the non recursive algorithm. Design an algorithm for element uniqueness in an array. Obtain its time complexity.	12	L3	CO2
	b.	List and explain the fundamental data structures with definition and examples.	08	L1	CO2
Module – 2					
Q.3	a.	Design a binary search algorithm and derive its time complexity and apply the same to search an element “42” from the given elements. 3, 14, 27, 31, 40, 42, 55, 66	12	L3	CO2
	b.	Sort the following elements using Quick Sort. Show only 1 st partition. 50, 30, 10, 90, 60, 40, 35, 62	08	L3	CO2
OR					
Q.4	a.	Define Topological Sorting. Obtain the topological ordering of elements using DFS and Source Removal Method, for the following graph Fig.Q4(a): Fig.Q4(a)	10	L3	CO2
	b.	Define Heap. Explain different types of heap. Sort the following elements using heap sort technique by creating bottom-up max heap tree. 26, 14, 18, 42, 6, 9, 38	10	L3	CO1
Module – 3					
Q.5	a.	Consider the following jobs with their profits and deadlines. Find the executing job sequence using greedy to obtain max profit. <p ₁ , p ₂ , p ₃ , p ₄ , p ₅ , p ₆ > = <23, 45, 6, 18, 60, 5> <d ₁ , d ₂ , d ₃ , d ₄ , d ₅ , d ₆ > = <3, 2, 1, 4, 2, 1>	10	L3	CO2

- b. Apply Kruskal algorithm to find Minimum Spanning tree for the below graph, Fig.Q5(b).

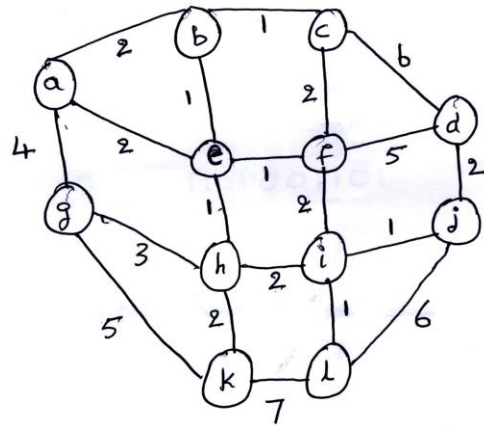


Fig.Q5(b)

OR

- Q.6 a. Find the shortest path from the vertex “d” to all other vertices for the below graph, Fig.Q6(a).

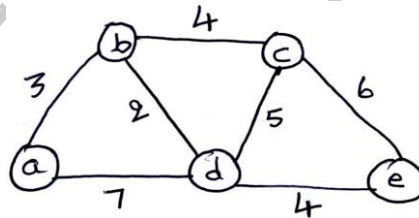


Fig.Q6(a)

- b. Construct a Huffman code for the following data:

Character	A	B	C	D	-
Prob	0.35	0.1	0.2	0.2	0.15

Find: (i) Huffman tree (ii) Decode the string “1001101101001101”

Module – 4

- Q.7 a. Apply Forward approach Multistage graph technique to find shortest path from S to t.

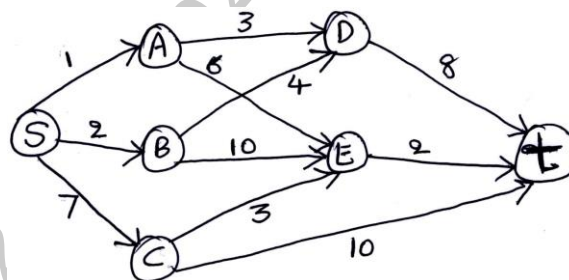


Fig.Q7(a)

- b. Find all pair shortest path for the below graph, Fig.Q7(b).

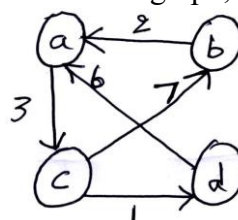
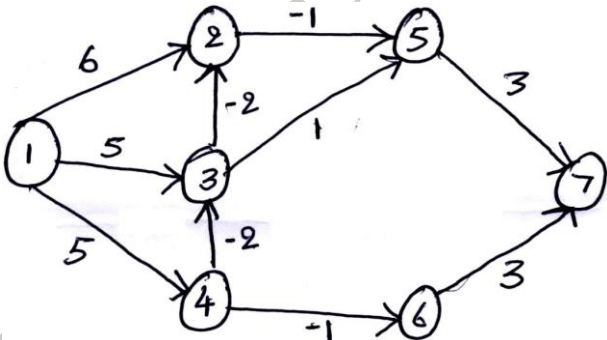


Fig.Q7(b)

OR

OR																				
Q.8	a.	Applying dynamic programming, find the optimal solution for the instance given below, with knapsack capacity $W = 5$. <table><tr><th>Item</th><th>Weight</th><th>Value</th></tr><tr><td>1</td><td>2</td><td>\$12</td></tr><tr><td>2</td><td>1</td><td>\$10</td></tr><tr><td>3</td><td>3</td><td>\$20</td></tr><tr><td>4</td><td>2</td><td>\$15</td></tr></table>	Item	Weight	Value	1	2	\$12	2	1	\$10	3	3	\$20	4	2	\$15	10	L3	CO3
Item	Weight	Value																		
1	2	\$12																		
2	1	\$10																		
3	3	\$20																		
4	2	\$15																		
	b.	Why we need Bellman-Ford algorithm? Apply the same to find the shortest path for the graph.  Fig.Q8(b)	10	L3	CO3															

Module – 5

Q.9	a. Define Backtracking. Find the solution space tree for 4-queens problem.	10	L2	CO1
	b. Find the subset from the given set with $d = 15$, $s = \{3, 7, 5, 6\}$ by constructing a state space tree.	10	L3	CO2

OR

Q.10	a. Solve the assignment problem using branch and bound technique. <table border="1" data-bbox="320 1245 603 1435"> <thead> <tr> <th></th><th>d_1</th><th>d_2</th><th>d_3</th><th>d_4</th></tr> </thead> <tbody> <tr> <td>p_1</td><td>9</td><td>2</td><td>7</td><td>8</td></tr> <tr> <td>p_2</td><td>6</td><td>4</td><td>3</td><td>7</td></tr> <tr> <td>p_3</td><td>5</td><td>8</td><td>1</td><td>8</td></tr> <tr> <td>p_4</td><td>7</td><td>6</td><td>9</td><td>4</td></tr> </tbody> </table>		d_1	d_2	d_3	d_4	p_1	9	2	7	8	p_2	6	4	3	7	p_3	5	8	1	8	p_4	7	6	9	4	10	L3	CO3
	d_1	d_2	d_3	d_4																									
p_1	9	2	7	8																									
p_2	6	4	3	7																									
p_3	5	8	1	8																									
p_4	7	6	9	4																									
	b. Solve the following travelling salesman problem using branch and bound technique by using the constraint “Visit City “b” before “c””. Fig.Q10(b)	10	L3	CO3																									

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Research Methodology and IPR

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Define Research. Elaborate on the objectives and significance of research.	10	L2	CO1
	b.	Explain the different types of research with an example for each.	10	L2	CO1
OR					
Q.2	a.	Distinguish between Research Method and Research Methodology.	10	L2	CO1
	b.	Illustrate the various steps involved in Research process with a neat diagram.	10	L2	CO1
Module – 2					
Q.3	a.	Define a research problem. Summarize the main techniques involved in defining a research problem.	10	L2	CO1
	b.	Explain the issues involved in defining a research and discuss on the necessity of defining the problem.	10	L2	CO1
OR					
Q.4	a.	Explain the literature review functions in research work and state the way in which it can help research.	10	L2	CO2
	b.	Illustrate the steps involved to conduct literature review for research and explain each one briefly	10	L2	CO2
Module – 3					
Q.5	a.	Discuss fully the important concept relating to research design.	10	L2	CO3
	b.	Briefly explain about importance of experimental design. Mention its types and explain any one in each type.	10	L2	CO3
OR					
Q.6	a.	Illustrate the steps that the researcher should pay attention while developing a sample design.	10	L2	CO3
	b.	The following are the number of departmental stores in 15 cities: 35, 17, 10, 32, 70, 28, 26, 19, 26, 66, 37, 44, 33, 29 and 28. If we want to select a sample of 10 stores, using cities as clusters and selecting within clusters proportional to size, how many stores from each city should be chosen? (Use a starting point of 10).	10	L3	CO3
Module – 4					
Q.7	a.	Clearly explain the differences between collection of data through questionnaire and schedules.	10	L2	CO4
	b.	Interpret the various steps involved in writing a report.	10	L3	CO4
OR					
Q.8	a.	What is interpretation? Discuss on the precautions that need to be taken by the research for correct interpretation.	10	L2	CO4
	b.	Elaborate on the different types of reports.	10	L2	CO4
Module – 5					
Q.9	a.	What is intellectual property? What are the different types of it?	10	L2	CO5
	b.	Explain briefly the Indian Patent Act 1970.	10	L2	CO5
OR					
Q.10	a.	Explain the importance of IPR in recent times.	10	L2	CO5
	b.	Discuss the salient features of Copyright Act 1957.	10	L2	CO5